Reg. No. | | | | | | | | | | | [14ME33]

THIRD SEMESTER B.TECH. EXAMINATION, DECEMBER - 2015

MATERIAL SCIENCE AND METALLURGY

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions choosing at least **TWO** questions from each part.
2. Draw neat sketches wherever necessary.
3. All questions carry equal marks.

PART-A

1.	a.	Explain with neat sketch the following: i. Frenkel. ii. Schottky defect.	07
	b.	State Fick's law of diffusion. Explain the various factors affecting diffusion.	07
2.		Explain plastic deformation of a single crystal by slip and twinning.	14
3.	a.	Explain the mechanism of solidification with a neat sketch.	07
	b.	Explain grain structure of cast metal and volumetric shrinkage in casting.	07
4.	a.	Explain stages involved in fatigue failure with a neat sketch.	08
	b.	Explain fatigue protection methods.	06

PART-B

5.		What is the purpose of case hardening? Classify the methods of case hardening and describe briefly any two of them.	14
6.	a.	Describe the classification of ferrous and non ferrous alloys.	07
	b.	Explain in detail the BIS and AISI classification of steels with examples.	07
7.		Write types of magnetism and explain any two of them.	14
8.	a.	What are the induced effects due to increase in surface area of Nano-particles?	07
	b.	Define Nano-composite. What are the types of Nano-composites?	07

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